

RMO(INDIA) – 2000

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. Let AC be a line segment in the plane and B a point between A and C. Construct isosceles triangles PAB and QBC on one side of the segment AC such that $m\angle APB = m\angle BQC = 120^\circ$ and an isosceles triangle RAC on the other side of AC such that $m\angle ARC = 120^\circ$. Show that PQR is an equilateral triangle.
 2. Solve the equation $y^3 = x^3 + 8x^2 - 6x + 8$, for positive integers x and y.
 3. Let $\langle x_1, x_2, \dots, x_n, \dots \rangle$ is a sequence of positive real number such that $x_1 \geq x_2 \geq \dots \geq x_n \geq \dots$, and for all n, $\frac{x_1}{1} + \frac{x_4}{2} + \frac{x_9}{3} + \dots + \frac{x_{n^2}}{n} \leq 1$. Show that for all k, the following inequality is satisfied:

$$\frac{x_1}{1} + \frac{x_2}{2} + \frac{x_3}{3} + \dots + \frac{x_k}{k} \leq 3.$$
 4. All the 7 digit numbers containing each of the digits 1, 2, 3, 4, 5, 6, 7 exactly once, one not divisible by 5, are arranged in the increasing order. Find the 2000-th number in this list.
 5. The internal bisector of angle A in a triangle ABC with $AC > AB$, meets the circum circle Γ of the triangle in D. Join D to the centre O of the circle Γ and suppose DO meets AC in E, possibly when extended. Given that BE is perpendicular to AD, show that AO is parallel to BD.
 6.
 - (a) Consider two positive integers a and b which are such that $a^a b^b$ is divisible by 2000. What is the least possible value of the product ab?
 - (b) Consider two positive integers a and b which are such that $a^b b^a$ is divisible by 2000. What is the least possible value of the product ab?
 7. Find all real values a for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$.

**